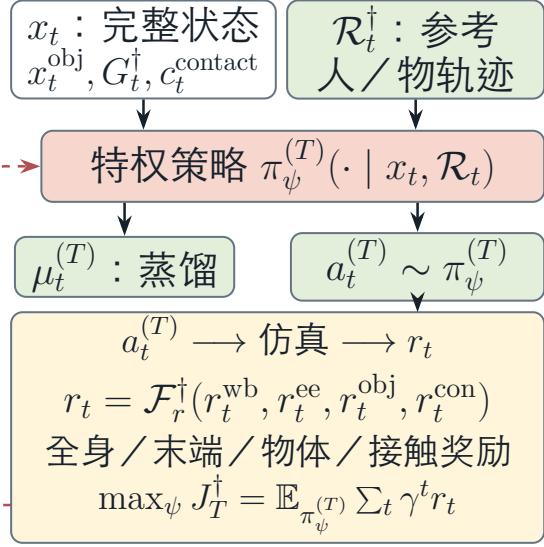


DreamMimic：三模型结构、训练与部署

实线：信息流 金色虚线：监督 红色虚线：更新 ξ ：世界模型参数 θ ：学生参数 $\psi^\dagger$ ：教师参数

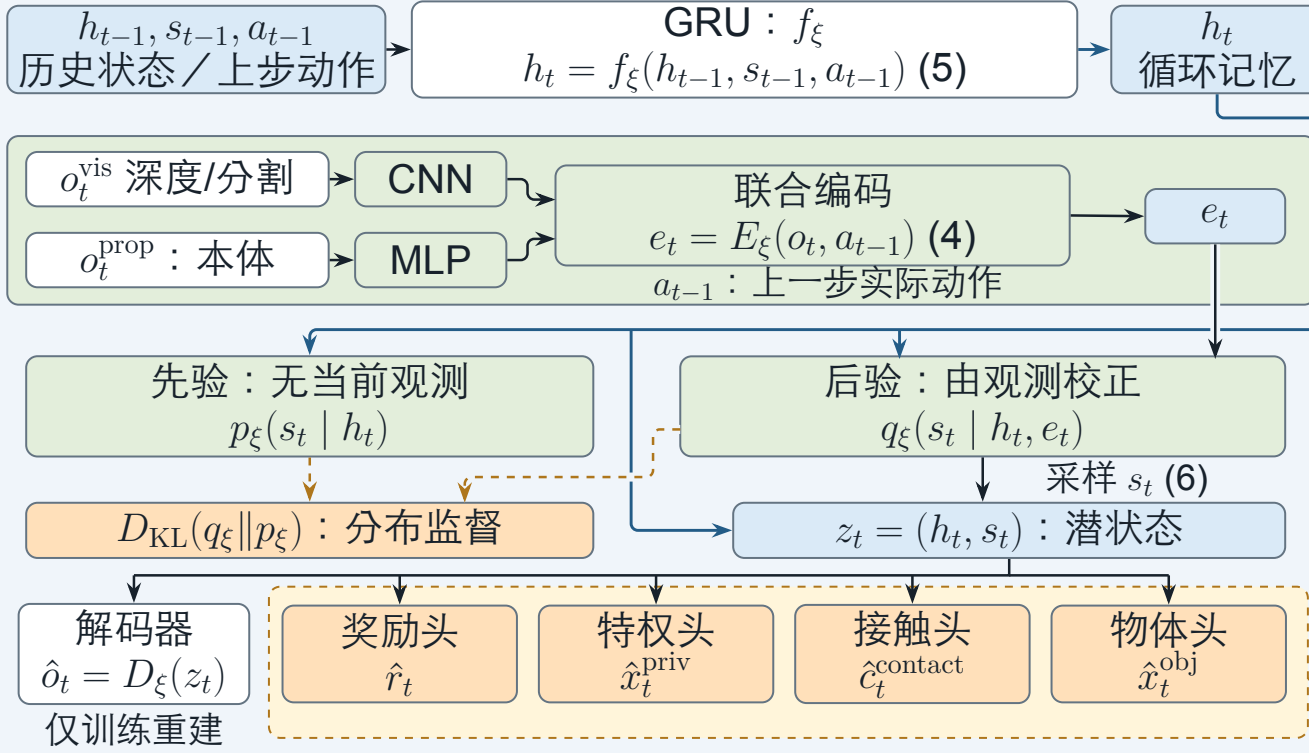
教师：RL，专用 $\rightarrow$ 通用



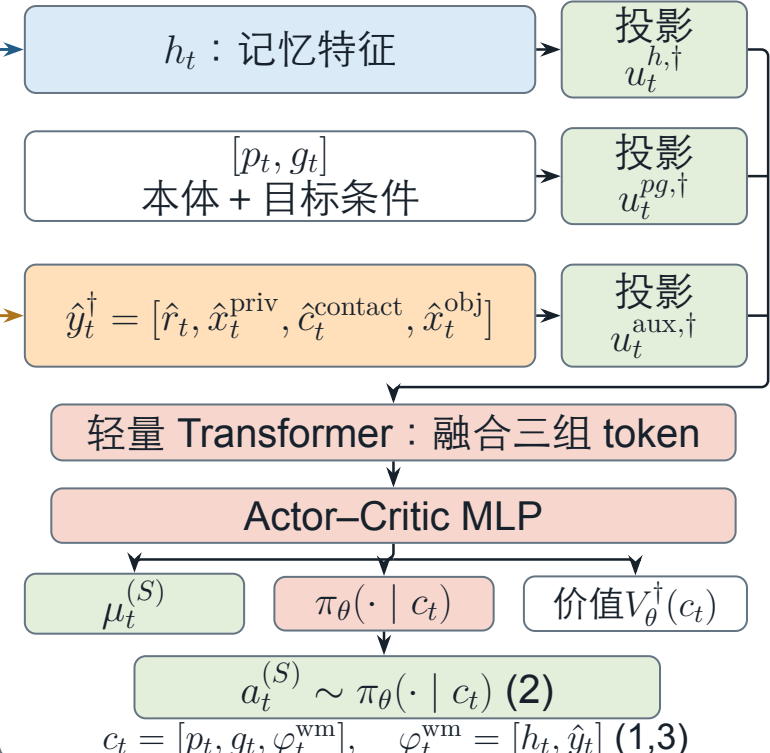
参考课程：重访困难片段

仅学生： $m = 0 \wedge \text{fail} \rightarrow \mathcal{B}_{\text{ref}}^\dagger$
 $\mathcal{B}_{\text{ref}} \rightarrow \mathcal{R}_{\text{init}}^\dagger \rightarrow \text{rollout}$

世界模型：记忆递推 + 观测校正 + 预测



学生：三组特征融合后出动作



世界模型损失：只更新 ξ

$$\mathcal{L}_{\text{wm}} = \mathcal{L}_{\text{KL}} + \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{rew}} + \mathcal{L}_{\text{priv}} + \mathcal{L}_{\text{con}} + \mathcal{L}_{\text{obj}} \quad (10)$$

$$Q_t^\dagger(s) = q_\xi(s | h_t, e_t), \quad P_t^\dagger(s) = p_\xi(s | h_t)$$

$$D_{\text{KL}}(Q_t || P_t) = \sum_s Q_t(s) \log \frac{Q_t(s)}{P_t(s)}$$

$$\mathcal{L}_{\text{KL}} = \alpha_{\text{dyn}} \mathcal{L}_{\text{dyn}} + \alpha_{\text{rep}} \mathcal{L}_{\text{rep}}$$

$$\mathcal{L}_{\text{dyn}}^\dagger = \mathbb{E}_{\mathcal{D}} \max(\tau, D_{\text{KL}}(\text{sg}[Q_t] || P_t))$$

$$\mathcal{L}_{\text{rep}}^\dagger = \mathbb{E}_{\mathcal{D}} \max(\tau, D_{\text{KL}}(Q_t || \text{sg}[P_t]))$$

$\dagger$ ：通用 stop-gradient / free-bits 展开

$$\mathcal{L}_{\text{rec}} = \text{MSE}(\hat{o}_t^{\text{vis}}, o_t^{\text{vis}}) + \text{MSE}(\text{symlog}(\hat{o}_t^{\text{prop}}), \text{symlog}(o_t^{\text{prop}}))$$

$$\text{MSE}(u, v) = \|u - v\|_2^2 / N, \quad N^\dagger: \text{元素数}$$

$$\text{symlog}(u) = \text{sign}(u) \log(1 + |u|)$$

$$\mathcal{L}_{\text{rew}} = \mathbb{E}_{\mathcal{D}} \ell_{\text{rew}}(\hat{r}_t, r_t)$$

$$\mathcal{L}_{\text{priv}} = \mathbb{E}_{\mathcal{D}} \ell_{\text{priv}}(\hat{x}_t^{\text{priv}}, x_t^{\text{priv}})$$

$$\mathcal{L}_{\text{con}} = \mathbb{E}_{\mathcal{D}} \ell_{\text{con}}(\hat{c}_t^{\text{contact}}, c_t^{\text{contact}})$$

$$\mathcal{L}_{\text{obj}} = \mathbb{E}_{\mathcal{D}} \ell_{\text{obj}}(\hat{x}_t^{\text{obj}}, x_t^{\text{obj}})$$

$$x_t^{\text{priv}} = [x_t^{\text{obj}}, G_t^\dagger, c_t^{\text{contact}}], \quad x_t^{\text{obj}} \in \mathbb{R}^{13}$$

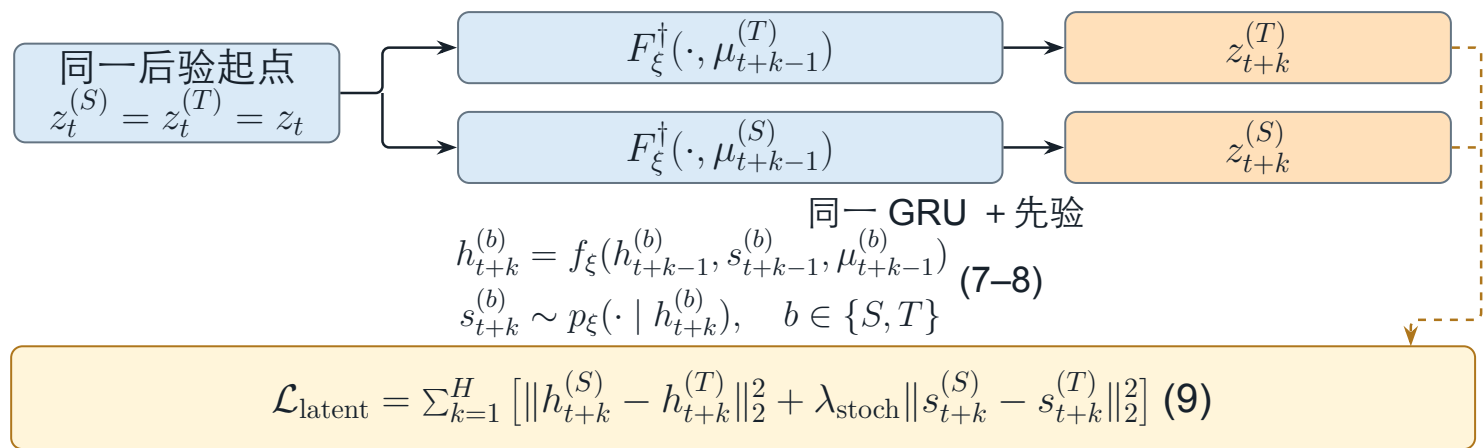
真值来自仿真/任务注释

$$h_t \in \mathbb{R}^{d_h}, \quad s_t \in \mathcal{S}, \quad a_t \in \mathbb{R}^{d_a}, \quad e_t \in \mathbb{R}^{d_e}, \quad z_t = (h_t, s_t)$$

$$p_t \in \mathbb{R}^{d_p}, \quad g_t \in \mathbb{R}^{d_g}, \quad x_t \in \mathbb{R}^{d_x}, \quad r_t \in \mathbb{R}$$

学生损失：固定教师与 WM，只更新 θ

$$\mathcal{L}_{\text{action}} = \|\mu_t^{(S)} - \mu_t^{(T)}\|_2^2$$



$H = 3$ ；不读未来观测；梯度经过 WM 回到学生动作

$$\mathcal{L}_{\text{total}} = c w_{\text{action}} \mathcal{L}_{\text{action}} + w_{\text{wm}} \mathcal{L}_{\text{latent}} + w_{\text{ppo}} \mathcal{L}_{\text{ppo}} \quad (11)$$

$$\kappa_t^\dagger = \pi_\theta(a_t | c_t) / \pi_{\text{old}}(a_t | c_t), \quad \hat{A}_t^\dagger: \text{优势}$$

$$\mathcal{L}_{\text{clip}}^\dagger = -\mathbb{E} \min(\kappa_t \hat{A}_t, \text{clip}(\kappa_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t)$$

$$\mathcal{L}_{\text{ppo}}^\dagger = \mathcal{L}_{\text{clip}}^\dagger + \lambda_V \mathbb{E}(V_\theta(c_t) - \hat{R}_t)^2 - \lambda_{\mathcal{H}} \mathcal{H}(\pi_\theta)$$

$$\mathcal{H}^\dagger(\pi_\theta) = -\mathbb{E}_{c_t} \mathbb{E}_{a \sim \pi_\theta(\cdot | c_t)} \log \pi_\theta(a | c_t)$$

$\hat{R}_t^\dagger$ ：回报目标； $\mathcal{H}^\dagger$ ：熵； $\epsilon^\dagger$ ：裁剪阈值。PPO 为通用展开

PCG：按真实表现分配教师/学生回合

$$m \sim \text{Bernoulli}(\rho)$$

$$\hat{r}_T \leftarrow \text{EMA}(r_t | m = 1)$$

$$\pi = \hat{r}_S / (\hat{r}_T + \varepsilon)$$

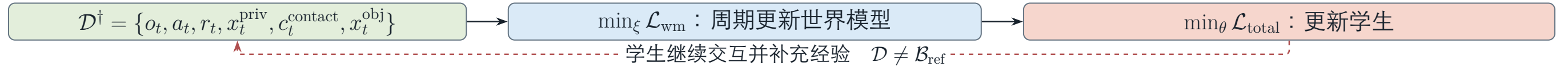
$$a_t = \begin{cases} a_t^{(T)}, & m = 1 \\ a_t^{(S)}, & m = 0 \end{cases}$$

$$\hat{r}_S \leftarrow \text{EMA}(r_t | m = 0)$$

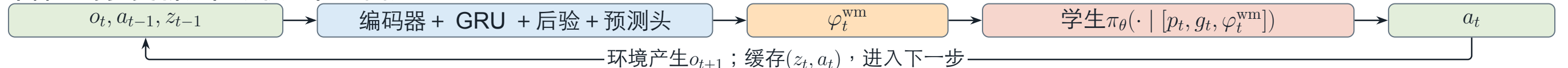
$$\rho_{\text{max}} \xrightarrow[\text{EMA}]{\pi \rightarrow \pi_{\text{target}}^\dagger} \rho_{\text{min}}$$

每回合选执行者； $c = \text{const}$ ；PPO 仅用学生回合，预热后低权重

交替训练：教师已固定，WM 与学生分别优化



部署：每步更新记忆；模型权重固定



p_t ：本体； g_t ：目标物体姿态 + 短时身体轨迹； $G_t^\dagger$ ：交互图； s_t ：离散随机态； $\hat{\cdot}$ ：预测； $\mu_t^{(T/S)}$ ：动作均值

$\dagger$ ：补充记号/通用展开